

CHARLOTTESVILLE-ALBEMARLE, VA, USA

WMO: 724016

Lat: 38.137N

Lon: 78.455W

Elev: 640

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 93736

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.6	(c) 19.2	(d) -5.1	(e) 4.3	(f) 22.3	(g) -0.1	(h) 5.6	(i) 26.0	(j) 18.8	(k) 44.6	(l) 17.4	(m) 44.0	(n) 3.8	(o) 310	(p) 0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.5	(c) 93.3	(d) 74.4	(e) 90.7	(f) 73.8	(g) 88.4	(h) 72.9	(i) 77.2	(j) 88.8	(k) 76.0	(l) 86.4	(m) 74.9	(n) 84.4	(o) 6.0	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.9	129.3	81.5	72.9	124.9	80.4	72.1	121.6	79.6	40.9	89.0	39.8	86.6	38.7	84.5	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.9	(b) 14.6	(c) 12.7	(e) 7.4	(f) 96.6	(g) 4.4	(h) 3.8	(i) 4.2	(j) 99.3	(k) 1.6	(l) 101.6	(m) -0.9	(n) 103.7	(o) -4.1	(p) 106.5		
(4) 16.9	(5) 14.6	(6) 12.7	(7) 7.4	(8) 96.6	(9) 4.4	(10) 3.8	(11) 4.2	(12) 99.3	(13) 1.6	(14) 101.6	(15) -0.9	(16) 103.7	(17) -4.1	(18) 106.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.3	36.6	39.6	47.0	57.2	65.5	73.3	76.8	75.4	68.8	58.0	48.0	40.2		
	DBStd	16.25	10.34	9.68	9.78	8.65	7.80	5.53	5.29	4.78	6.64	8.22	8.58	8.97		
	HDD50	1441	434	314	179	28	1	0	0	0	0	22	137	326		
	HDD65	4079	882	711	560	263	92	7	1	1	37	246	511	768		
	CDD50	4107	17	23	86	244	483	699	832	788	564	270	78	23		
	CDD65	1271	0	0	3	29	109	256	368	324	151	29	2	0		
	CDH74	11451	1	5	47	361	989	2274	3513	2809	1176	257	17	2		
	CDH80	4273	0	0	4	92	297	824	1517	1093	384	61	1	0		
(14) Wind	WSAvg	4.5	5.4	5.4	5.8	5.7	4.6	4.0	3.5	3.3	3.6	4.0	4.5	4.5		
(15) Precipitation	PrecAvg	42.5	3.0	2.4	3.8	3.0	4.1	3.9	4.4	3.3	4.5	3.7	3.1	3.4		
	PrecMax	68.0	6.6	7.3	8.3	5.0	8.7	7.0	10.0	6.0	9.7	9.1	6.6	7.4		
	PrecMin	31.3	1.1	0.5	0.2	1.1	1.1	0.7	1.2	0.9	0.6	0.0	0.4	1.9		
	PrecStd	8.9	1.7	1.7	1.8	1.1	2.2	1.8	2.4	1.2	3.0	2.6	1.9	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.0	73.1	79.7	87.2	90.3	94.6	98.5	95.0	93.3	86.2	76.1	70.7		
		MCWB	57.8	59.3	61.1	64.9	72.1	74.7	76.2	74.7	73.1	70.9	62.2	58.4		
	2%	DB	63.0	65.8	73.7	82.4	86.6	90.5	94.2	91.7	88.1	80.6	70.5	64.0		
		MCWB	54.4	53.4	57.9	63.1	70.3	73.1	75.1	74.3	72.1	66.5	57.6	56.2		
	5%	DB	57.9	60.9	68.8	78.6	83.4	88.1	91.0	89.6	84.2	76.1	66.3	59.3		
		MCWB	50.4	49.0	54.8	61.1	68.5	72.5	74.5	73.9	70.1	64.9	56.3	52.6		
	10%	DB	52.8	55.1	64.1	73.4	80.0	84.9	88.2	86.3	81.0	72.3	63.0	54.5		
		MCWB	45.5	45.9	52.0	58.8	66.8	71.0	73.6	72.5	68.6	62.7	54.6	47.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.0	61.7	63.4	69.1	74.8	77.2	78.7	78.6	76.2	73.6	65.1	63.5		
		MCDB	65.0	69.7	73.1	78.7	85.3	88.6	91.4	89.6	87.1	81.9	70.9	67.3		
	2%	WB	56.6	55.7	60.4	66.0	72.6	75.5	77.5	77.0	74.5	70.7	62.2	58.1		
		MCDB	62.0	63.1	70.4	77.9	82.9	86.2	89.8	87.6	83.3	77.0	67.8	63.0		
	5%	WB	51.2	51.0	57.3	63.4	70.7	74.4	76.3	75.6	72.8	67.8	58.7	53.1		
		MCDB	56.3	57.9	66.2	74.2	80.2	84.2	87.3	84.6	80.1	73.7	64.7	57.9		
	10%	WB	46.0	46.4	53.8	61.0	68.9	73.1	75.1	74.3	71.2	64.2	55.3	48.1		
		MCDB	52.0	54.6	62.2	70.9	77.6	81.9	84.9	82.6	77.6	69.8	60.8	53.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	18.7	20.5	22.0	20.5	19.8	19.5	19.3	19.7	20.6	19.9	17.0		
		MCDBR	23.5	26.7	26.7	28.7	24.4	23.1	22.6	22.6	24.0	25.4	25.4	22.4		
		MCWBR	17.2	17.3	15.2	13.4	11.2	9.4	8.1	8.5	10.0	13.0	15.6	15.8		
	5% WB	MCDBR	20.7	23.2	23.6	23.6	21.0	20.3	19.8	19.2	18.4	19.2	19.6	19.4		
		MCWBR	17.0	17.1	15.0	12.8	10.4	9.1	7.9	8.4	8.8	11.3	14.6	16.0		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.315	0.349	0.393	0.437	0.484	0.505	0.512	0.408	0.354	0.319	0.300		
	taud		2.509	2.442	2.396	2.299	2.196	2.104	2.071	2.027	2.307	2.453	2.509	2.531		
	Ebn at Noon		283	290	289	281	269	256	249	244	266	272	272	274		
	Edh at Noon		25	30	35	41	46	50	52	53	38	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		709	961	1276	1592	1767	1945	1856	1673	1385	1081	795	622		
	RadStd		56	109	77	123	152	127	107	82	162	123	59	53		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.84	N/A	N/A	N/A	+0.72	N/A	N/A	N/A	+580	+392
(47) Regional (55 neighbors)		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	+131

Nomenclature: See separate page